

Human Development Index (HDI) Prediction using Machine Learning

UAT Execution & Report Submission for the HDI Prediction Flask Web Application

Date	04 July 2026
Team ID	
Project Name	A comprehensive measure of well being
Maximum Marks	4 Marks

1. Purpose of Document

The purpose of this document is to briefly explain the test coverage and open issues of the **Human Development Index (HDI) Prediction using Machine Learning** project at the time of release to User Acceptance Testing (UAT). It summarises defects identified across the Flask-based prediction web application and the underlying Random Forest classification model, along with the status of test cases executed across the system's functional modules.

2. Defect Analysis

This report shows the number of resolved or closed bugs at each severity level, and how they were resolved.

Resolution	Severity 1	Severity 2	Severity 3	Severity 4	Subtotal
By Design	1	1	2	3	7
Duplicate	0	0	1	2	3
External	0	1	1	1	3
Fixed	3	4	6	9	22
Not Reproduced	0	0	1	1	2
Skipped	0	0	0	1	1
Won't Fix	0	0	1	2	3
Totals	4	6	12	19	41

Total defects logged: 41. Of these, 22 were fixed prior to UAT release, with the remaining categorised as by-design behaviour, duplicates, external dependencies, or minor issues that were skipped or will not be fixed in this release, none of which affect core prediction functionality.

3. Test Case Analysis

This report shows the number of test cases that have passed, failed, and untested, grouped by functional area of the HDI Prediction application.

Section	Total Cases	Not Tested	Fail	Pass
Data Preprocessing & Validation	8	0	0	8
Model Prediction Engine	10	0	0	10
Flask Web Application (UI)	9	0	0	9
Input Handling & Error Messages	6	0	0	6
Performance & Response Latency	4	0	0	4
Cross-Browser Compatibility	3	0	0	3
Totals	40	0	0	40

All 40 test cases executed across the six functional areas of the HDI Prediction application passed successfully, with zero failures and zero untested cases, consistent with the functional and performance testing results recorded in the project's Final Report (Section 6).